DAEN/ISEN 427 In class assignment 3

Due: Tue Oct 7 10:50 AM [*10 pts*]

M&Ms – Working Sheet

Name/s: ______________________________ **Date:** __________

1. Setup, prior choice **[*1 pt*]**

Pick a color. Define θ as the long-run proportion of that color in a bag.

- **Chosen color:** __________
- **Prior:** $\theta \sim \text{Beta}(\alpha_0, \beta_0)$ with $\alpha_0 =$ __________ and $\beta_0 =$ __________

Justify your prior choice (what beliefs/information does it encode):

2. Data collection (round 1) **[*2 pts*]**

Let x_1 be the count of your color and n_1 the total number of M&Ms in your bag.

- $n_1 =$ __________ $x_1 =$ __________

Please record the number of M&Ms for each color and total and share with your colleagues before eating it!

yellow __________, *red* __________, *orange* __________, *brown* __________, *blue* __________, *green* __________

Posterior after Round 1. Using Beta-Binomial conjugacy:

$$\theta \mid x_1 \sim \text{Beta}(\alpha_1, \beta_1), \quad \alpha_1 = \alpha_0 + x_1, \quad \beta_1 = \beta_0 + n_1 - x_1.$$

Compute:

- $\alpha_1 =$ __________
- $\beta_1 =$ __________
- Posterior mean $\mathbb{E}[\theta \mid x_1] = \dfrac{\alpha_1}{\alpha_1 + \beta_1} =$ __________
- 95% credible interval (CI) for θ: __________ __________

Sketch the prior and posterior shape:

3. Sequential updating (rounds 2 & 3) **[*2 pts*]**

Repeat two more samples of size n_2 and n_3 (ask the people in your table).

	n_t	x_t	α_t	β_t	Mean $\frac{\alpha_t}{\alpha_t+\beta_t}$	95% CI
Prior (t=0)	–	–	$\alpha_0 =$ ______	$\beta_0 =$ ______	______	–
Round 1 (t=1)	______	$x_1 =$ ______	$\alpha_1 =$ ______	$\beta_1 =$ ______	______	____________
Round 2 (t=2)	______	$x_2 =$ ______	$\alpha_2 =$ ______	$\beta_2 =$ ______	______	____________
Round 3 (t=3)	______	$x_3 =$ ______	$\alpha_3 =$ ______	$\beta_3 =$ ______	______	____________

Note: Update rules each round: $\alpha_t \leftarrow \alpha_{t-1} + x_t$, $\beta_t \leftarrow \beta_{t-1} + n_t - x_t$.

4. Posterior predictive check **[*1 pt*]**

Use your latest posterior $\theta \sim$ Beta(α_3, β_3) to predict the number of your color in a future bag of size m.

a) Give the posterior predictive distribution of Y (number of your color in a bag):

$$Y \mid \text{data} \sim \text{Beta-Binomial}\left(n = m,\ \alpha = \alpha_3,\ \beta = \beta_3\right).$$

b) Compute the posterior predictive mean $\mathbb{E}[Y \mid \text{data}] = m \cdot \dfrac{\alpha_3}{\alpha_3 + \beta_3} =$ ________

c) Give a 95% predictive interval for Y: ________ ________

d) Check this value using a 5^{th} bag (in your table or in a nearby table).

5. Sensitivity to the prior **[*2 pts*]**

Suppose you had instead chosen one of the following priors.

Uniform prior: Beta$(1, 1)$ Skeptical prior: Beta$(2, 8)$ Optimistic prior: Beta$(8, 2)$

Without recomputing everything from scratch, explain qualitatively (and compute one round if time permits) how your posterior would differ after the same data.

6. Interpretation, assumptions **[*2 pts*]**

a) Interpret your final posterior mean and 95% CI for θ. What does it say about the true color proportion?

b) List the modeling assumptions you made (e.g., independence, identical draws, well-mixed bags) and discuss whether they are reasonable here.